

Final Lab Exam – Computer Networks Lab (Spring 2025) – Syllabus & Important Instructions

Dear Students,

Please find attached the detailed syllabus for the Final Lab Exam of Computer Networks Lab – Spring 2025.

It should be clearly noted that all labs from Lab # 01 to Lab #12 are included in the final lab exam. No lab is excluded, and each carries equal importance.

This PDF file also contains resources and reference material for each lab, which may assist you in your preparation.

Important Details:

- The Final Lab Exam will begin from 5th May 2025, following your usual class schedule (day and time).
- As per the directive of the respected HOS, there will be no makeup/retake under any circumstances. If you miss the exam, you may only appear in another section of the same course (if available).
- The final exam will consist of 04 questions, and the time duration will be more than 2 hours, that will be mentioned on the question paper.
- Submission must be completed within the allocated time. **No late submissions will be accepted via email, USB, or any other medium.** Failure to submit on time will result in 0 marks.

Please prepare accordingly and ensure your punctuality and presence on your respective lab exam day.

Detailed Course Breakdown

LAB # 01

1. Definitions: Network, Computer Network, Types of Computer Network, Host, Topology, RJ45, Types of Cables (Straight Cable & Crossover Cable).
2. Network Terminologies: NIC, MAC, Hub, Switch, Router, IP address, Port address, Gateway address, Domain Name, DNS server, DHCP.
3. OSI Model.
4. Commonly Used Commands in Networking: ipconfig, hostname, nslookup, ping, tracert, netstat.
5. IP Addressing (Below mentioned topics were covered):
 - Classes of IP (A, B, C, D, and E),
 - Private Address Space,
 - Default Subnet Masks,
 - Network Address Identification,
 - Host Address Identification.

LAB # 02

1. Introduction to Packet Tracer
2. Concept of DHCP and its working in Cisco Packet Tracer

LAB # 03

1. Basic Introduction of Socket Programming (Theory).
2. Socket Programming in Python along with its useful functions.
3. Server Socket Methods (bind, listen, accept).
4. Client Socket Methods (connect, recv, send, recvfrom, sendto, close, socket.gethostname()).
5. Getting IP address in Python.
6. Getting any host IP address in Python.
7. Getting hostname by IP address in Python.
8. Getting Service Name, Given Port Number & Protocol in Python.
9. Port Scanner.
10. Client Server Connecting Programs.

LAB # 04

1. HTTP

2. HTTPS
3. Client Errors
4. Server Errors
5. Implementation for HTTP & HTTPS in Packet Tracer.
6. Displaying headers for HTTP & HTTPS.
7. DNS in Packet Tracer
8. Record and Canonical Record.
9. Implementation of DNS.
10. Implementation of DNS with multiple servers (with switch and with router).

LAB # 05

1. SMTP (with single and multiple servers using switch and also routers).
2. FTP.
3. Wireshark.

LAB # 06

1. Introduction to Telnet & configuration of Telnet in Cisco Packet Tracer.
2. Introduction to SSH & configuration of SSH in Cisco Packet Tracer.
3. Introduction to ACL and configuration of ACL in Cisco Packet Tracer (Standard ACL + Extended ACL).

LAB # 07

1. Introduction to NS3 network simulator.
2. Simple network simulation in NS3.
3. Implementation of TCP congestion control & simulation in NS3.

LAB # 08

1. Introduction to Wireless Network.
2. Importance of Wireless Technology.
3. Working of Wireless Network.
4. Configuration of Wireless Network.
5. Introduction to NAT & its Types (Static NAT & Dynamic NAT).
6. Configuration of NAT.

LAB # 09

1. Understand Software Defined Networks.
2. Understand what Mininet is and why it is useful for testing network topologies.
3. Invoke Mininet from the CLI.
4. Testing connectivity.
5. Mininet CLI commands.

LAB # 10

1. Introduction to Subnets & Subnetting.

2. Purpose of Subnetting.
3. Subnet tables of different IPv4 classes.
4. Introduction of CIDR.
5. Implementation of Subnetting.
6. Introduction to Static Routing and its implementation.
7. Introduction to Routing protocol & Routed protocol.
8. Classes of Routing Protocols.
9. Types of Dynamic Routing & its configuration (RIP).

LAB # 11

1. Types of Dynamic Routing & its configuration (OSPF).
2. Types of Dynamic Routing & its configuration (BGP).

LAB # 12

1. Introduction to Virtual Area Networks - VLANs.
2. Types of Connections in VLAN (access and trunks).
3. Trunks with routers and with switches.
4. Introduction to InterVLAN routing.
5. Configuration of VLAN.
6. Configuration of InterVLAN routing.

RESOURCES:

LAB # 01

OSI MODEL

https://youtu.be/vv4y_uOneC0?si=9RWeenGVi3q0oCi

Find Range, Network Id, Host, Broadcast address with Numerical Examples

https://youtu.be/vTzrn_M77mo?si=PBlpoRwMDBpGiTBG

<https://youtu.be/q7wNcYliJ1Q?si=B1zs3a6F5BxGj5L2>

IP Addressing

https://youtu.be/_ISu9f8ofZk?si=6iunnnNeSL95hrNu

LAB # 02

DHCP

<https://youtu.be/e6-TaH5bkjo?si=AVqiNsBc1rvxNgV7>

<https://youtu.be/4pkDL1pgCgQ?si=L1i2LfopN5W2dfn5>

DHCP working in cisco packet tracer

https://youtu.be/ZCz9rRhzy4?si=idm82OPZt_Tp43-A

How to Configure DHCP in CISCO router in Packet Tracer (Additional)

https://youtu.be/DsL_JEdMPXg?si=x4RpuGmW8UuFne-W

LAB # 03

Complete playlist of Socket Programming

<https://www.youtube.com/watch?v=l5WU7d49OGk&list=PLS1QulWo1RIZGSgRsn0b8w9uoWM1gHDpo>

LAB # 04

http / https

<https://youtu.be/R4OcAG-orLg?si=X3sh0h8-3QmeyHZZ>

How to configure HTTP service

<https://youtu.be/eN4G7cyyoSo?si=G-YrnPEmVvnFTd9G>

DNS and its records

<https://youtu.be/HnUDtycXSNE?si=zXKIgyD9rN74xfy7>

<https://youtu.be/FhC62Lm2338?si=9RrFHR3OLUU6bza7>

How to configure DNS Server

<https://youtu.be/CyEacSDD0V4?si=CyamE7OHETwVbMcE>

http status codes

<https://youtu.be/LesZgTGP-LM?si=WOxuqgWAxw2vGL25>

LAB # 05

SMTP

<https://youtu.be/PJo5yOtu7o8?si=BJ2pSPtZ-qTO2FdJ>

How to configure Email Server | Email Server configuration step by step in cisco packet tracer

https://youtu.be/y7v9EJyXo-Q?si=Cywl2uuh_711dTho

FTP

https://youtu.be/tOj8MSEIbfA?si=g9lLZ1HQk6r_9XEw

FTP Server Using CISCO Packet Tracer

https://youtu.be/Mk5WUsHOK0Y?si=L9pCCUWXwDcrPH_9

<https://youtu.be/vCCBEZxBjQI?si=FoFjT-sgWIBurwhz>

Wireshark

Complete playlist of basic wireshark

https://www.youtube.com/playlist?list=PLZUlpNYznmNTL02BhILupSz7pqngYw_uV

LAB # 06

Telnet vs SSH

<https://youtu.be/tZop-zjYkrU?si=rMuX6AjNMPES6E-h>

SSH KEYS

<https://youtu.be/dPAw4opzN9g?si=PKZoWhro4lneW45t>

Telnet Configuration in Cisco Packet Tracer

<https://youtu.be/ADB30TSBUHE?si=hjq2cUsS9SOWDk66>

Telnet on Router in Cisco Packet Tracer

<https://youtu.be/lpBbikiUUuU?si=tTndnj8LS8wqCHrv>

Telnet on Switch in Cisco Packet Tracer

<https://youtu.be/hP7mrNW9HFo?si=eDO1dvZGy9jVshgV>

Packet Tracer Router & Switch SSH Configuration

https://youtu.be/h7BH2tSAmwU?si=YJ4RbRY35pf_ytKL

https://youtu.be/W7VzCGezB90?si=_a9ekRmBwBrcbqeg

LAB # 07

Complete ACL

https://www.youtube.com/watch?v=pj6A1bbWaOo&list=PLCaLsjNlEvhFMMqk5NJHHCM3I42BP7_Lo

Standard ACL Configuration Packet Tracer

https://youtu.be/Bl6dZJq_Wc0?si=C1c6xOhhOXkU_uFC

Extended ACL Configuration Packet Tracer

https://youtu.be/6qfxl8jLbUg?si=aJwqG7HTBd_gA5UI

Configuring the VTY Lines with Access Control List

<https://youtu.be/gynnFvfEDhM?si=Up7V7aLxAhD1rUrW>

LAB # 08

NS3

<https://mail.google.com/mail/u/0/?tab=rm&ogbl#search/NS3/FMfcgzQZTVsdzpFLHcKGZQvpzXXDvwpr?projector=1>

LAB # 09

Wireless router configuration in cisco packet tracer

https://youtu.be/9B1UQq_309k?si=Zng8I6bzuM0DD8Hx

concept of Nat and its types

<https://youtu.be/yEhg5gTVCAE?si=dYUt13kqmLNGcpg8>

Configuring Static NAT on Cisco Router

<https://youtu.be/Q86HzPdphs?si=L1Y4f3WUWmRgigL9>

Configuring Dynamic NAT on Cisco Router|

https://youtu.be/cZX9wRcV_Rs?si=cXAF5V19we9QI_LQ

Configuring PAT or NAT Overload on a Cisco Router

<https://youtu.be/NSn3qNToPa4?si=qI3O4WBkNXp9TVEr>

LAB # 09

Basic Mininet commands

https://youtu.be/StxQtQ_2Yk0?si=GTt9Vm4bE2YeY1DK

LAB # 10

Subnetting and its problems

<https://www.youtube.com/@networkingplus>

<https://youtu.be/DqU3KVCyFNg?si=MP-USugqeBXq2RxS>

https://youtu.be/NuVqcOq9YtY?si=WBJFu0o2lh8_sc-i

<https://youtu.be/0sBEJiUoTPI?si=cAcDaCZIx6Jr6yJe>

Subnetting Practice Example: Class C Address (Simulation in Cisco Packet Tracer)

<https://youtu.be/HsBJG2tIWmk?si=3A1hxdAyQo7TonEX>

Static routing configuration

<https://youtu.be/lmnptnqn-WI?si=PGDOWIuArYVQq3Qb>

LAB # 11

RIP

https://youtu.be/CTm6FAK4oO8?si=sk3_54r1r72MPqbE

OSPF

https://youtu.be/3SHtTW3EFuc?si=_08hKoN9qQ5SIBVa

BGP

<https://youtu.be/Y71DK9pP8u0?si=ChDwLKhaNhLEntOQ>

LAB # 12

VLAN

<https://youtu.be/kOLeiXpA-5I?si=wiYmwVZJRRlIbTx>

Inter-vlan

<https://youtu.be/3II2RwiXImg?si=AkAcFbW26WeDmB9h>

<https://youtu.be/apwWzXjoVXE?si=aT5vM6jxXeNIcd6p>

Wishing you the best of luck for your final lab exam!

Best regards,

Muhammad Nadeem Ghouri